Inter Planetary Weights Using Dictionaries and Pickling

Important AI Statement: Students are strictly prohibited from using AI tools (ChatGPT, Copilot, or any other code-generating software) to complete this assignment. All code must be written manually by the student. Any detected AI-generated code will result in a failing grade for that assignment.

- **Requirement:** Submit only your own original code. Do NOT copy, paste, or modify code generated by AI. You may refer to textbooks, class notes, or personal experimentation only.
- **Honor Statement:** By submitting this assignment, you affirm that all code is your own work and was not created using AI code generation tools.

Penalty Statement: Violations of this policy will be treated as academic dishonesty, subject to the institution's academic integrity rules and 0 for this assignment.

This is a throwback to one of the first Python programs you wrote in a previous class. You can calculate a person's weight on the different planets within our solar system by multiplying their mass by the gravity factor on the surface of the planet.

Weight = Earth Weight x Surface Gravity Factor

Planet	Surface Gravity Factor
Mercury	0.38
Venus	0.91
Moon	0.165
Mars	0.38
Jupiter	2.34
Saturn	0.93
Uranus	0.92
Neptune	1.12
Pluto	0.066

For example if you weigh 100 pounds on earth on Mars you would be:

Mars Weight 38 = 100 x .38

You Must Code the following:

1. You will need to code a main() function.
2. Create a dictionary that contains the conversion factors for each planet. You will need to figure out the **Key** and the **Value** portions.

3. Open up the Pickling file that stored previous values when the program was run in the past.
    - Name the file with your initials and PlanetaryWeights such as **bcPlanetaryWeights.db**
    - Code a try/except to handle if the file does not exist
    - Transfer the contents of the .db into a dictionary called **dictPlanetHistory**
4. Prompt the user if they wish to see the history with a **Y** or **y** or **N** or **n**. If **y** print out the previous entries in the dictionary **dictPlanetHistory**
5. Code a loop that does the following:
    - Prompt the user for a unique Name you will have to handle upper and lower case spelling to make sure the same name does not exist more than once. Check if the name is already in the dictionary **dictPlanetHistory**. If it is already there ask for another unique name.
    - A blank name entered will should exit the loop
    - Prompt for Earth Weight. Make sure you have try/except to handle invalid numeric data and keep prompting until a valid number isentered.
    - Declare another dictionary called **dictPersonWeights**
    - Code another loop:
        - Use the Surface Gravity Factors Planetary Weights dictionary to compute and output the Earth Weight for each of the planet's Surface Gravity Factors
        - For each planet add the Planet Name and the computed Planet Weight to the dictionary **dictPersonWeights**
    - Add the Person's Name and the **dictPersonWeights** to the history dictionary **dictPlanetHistory**
6. When you exit the loop from step 5 and before the main() function ends pickle/output the **dictPlanetHistory** to the db file created in step 3.
7. Refer to the sample output below and make sure you include the following requirements and you **MUST use Python's f-strings syntax** we covered in a previous class.
    - Output the inputted name and make sure the words Solar System's includes the single quote
    - Make sure all the computed planet weights are lined up exactly in the sample output
    - The outputted weight needs to take up 10 positions with 2 decimal points.
    - Test with 70.5 to make sure code works. Then try your own weight.
8. Use comments to document your code. At the top of your code place a Comment Heading Block that has these elements: Your Name, Code Name and you must answer these reflection questions:
    - Share what did you liked about this assignment?
    - Share what you struggled with?
    - How did you like working with dictionaries and pickling concepts?
    - Share your top 2 things you learned on this assignment.

```
# Name: YourName
# Assignment: Planetary Weights Dictionaries
# Reflection:  Share what you liked about this assignment?
#              Share what you struggled with?
#              How did you like working with dictionaries and pickling concepts?
#              Share exactly 2 things you learned on this assignment:
#               1.
#               2.
```

9. Put s prefixes in front of variables that contain string values and **i** prefix for variables that contain integer numbers and **f** for floating numbers. Add your initials such as BLC at the end of each variable (not constants). For example: fWeightEnteredBLC.
10. This assignment is for learning purposes you cannot use AI/ChatGPT to create the code.

Sample Output:

```
What is your name (enter key to quit): Moose
What is your weight: 70.5
Moose, here are your weights on our Solar System's planets.
Weight on Mercury:          26.79
Weight on Venus:            64.16
Weight on our Moon:         11.63
Weight on Mars:             26.79
Weight on Jupiter:         164.97
Weight on Saturn:           65.56
Weight on Uranus:           64.86
Weight on Neptune:          78.96
Weight on Pluto:             4.65
What is your name (enter key to quit):
```

Output sample with History: Red box is the history and Green box is the new computation:

```
Would you like to see the history y/n: y
Moose, here are your weights on our Solar System's planets.
Weight on Mercury:          26.79
Weight on Venus:            64.16
Weight on our Moon:         11.63
Weight on Mars:             26.79
Weight on Jupiter:         164.97
Weight on Saturn:           65.56
Weight on Uranus:           64.86
Weight on Neptune:          78.96
Weight on Pluto:             4.65
What is your name (enter key to quit): bentley-louis
What is your weight: 49.5
Bentley-Louis, here are your weights on our Solar System's planets.
Weight on Mercury:          18.81
Weight on Venus:            45.05
Weight on our Moon:          8.17
Weight on Mars:             18.81
Weight on Jupiter:         115.83
Weight on Saturn:           46.04
Weight on Uranus:           45.54
Weight on Neptune:          55.44
Weight on Pluto:             3.27
What is your name (enter key to quit):
```

```
Would you like to see the history y/n: n
What is your name (enter key to quit): Moose
Moose is already in the history file.  Enter an unique name.
What is your name (enter key to quit):
```

Grading Rubric

Criteria	Meets (100%)	Somewhat (50%)	Not Present (0%)
dictPlanetHistory dictionary is created 20 points	A dictionary is present and coded as specified in this assignment.	A dictionary is present but not coded as specified in this assignment.	A Dictionary is not present.
Calculations 10 points	Calculations were done properly with the correct results.	Calculations were attempted but with some errors or incorrect results.	Calculations were not attempted.
Output Formatting 10 points	Output was correctly formatted using f-strings per the sample.	Output was attempted using f-strings but did not match the sample.	No f-strings formatting was attempted.
Pickling file dictionary with input/output functions 35 points	Pickling was created for both input/output and is coded correctly.	Pickling was attempted for either input or output and is coded correctly.	Pickling was attempted.
Comment Block Reflections, Comments and Variable Prefixing and Suffixing 25 points	Comment Heading Block is present and reflection responses are present. Comments present and reflection questions are fully answered and variable prefixing and initial suffixing.	Either Comment Heading Block/Comments present but answered vaguely or copied or Variable prefixing/suffixing is present	No Comment Heading Block/Comments present and No variable prefixing/suffixing.